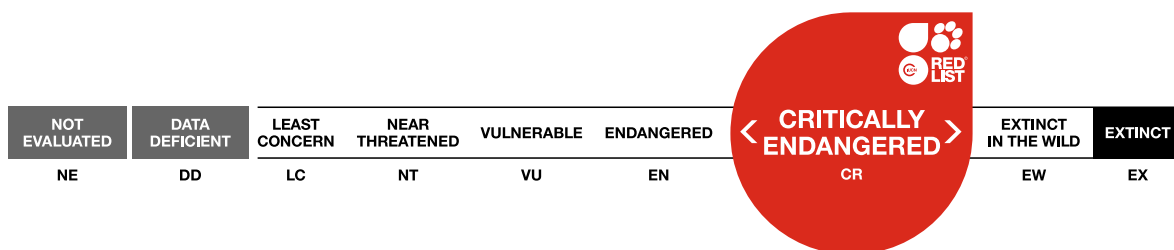


Calotriton arnoldi, Montseny Brook Newt

Assessment by: IUCN SSC Amphibian Specialist Group



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Amphibia	Caudata	Salamandridae

Scientific Name: *Calotriton arnoldi* Carranza & Amat, 2005

Common Name(s):

- English: Montseny Brook Newt, Montseny Newt

Taxonomic Source(s):

Frost, D.R. 2021. Amphibian Species of the World: an Online Reference. Version 6.1. New York, USA. Available at: <https://amphibiansoftheworld.amnh.org>.

Assessment Information

Red List Category & Criteria: Critically Endangered A3ce; E [ver 3.1](#)

Year Published: 2022

Date Assessed: August 20, 2021

Justification:

Listed as Critically Endangered due to a projected population decline, estimated to be more than 80% over the next 9–18 years (which represents three generation lengths), from the potential impact of *Batrachochytrium salamandrivorans* (*Bsal*) on this species and from a deterioration in habitat quality as a result of water abstraction. Furthermore, extinction risk modelling for *Bsal* indicates that the probability of extinction for this species will be >50% in the next three generation lengths (9–18 years).

For further information about this species, see [Supplementary Material](#).

Previously Published Red List Assessments

2009 – Critically Endangered (CR)
<https://dx.doi.org/10.2305/IUCN.UK.2009.RLTS.T136131A4246722.en>

2008 – Critically Endangered (CR)

Geographic Range

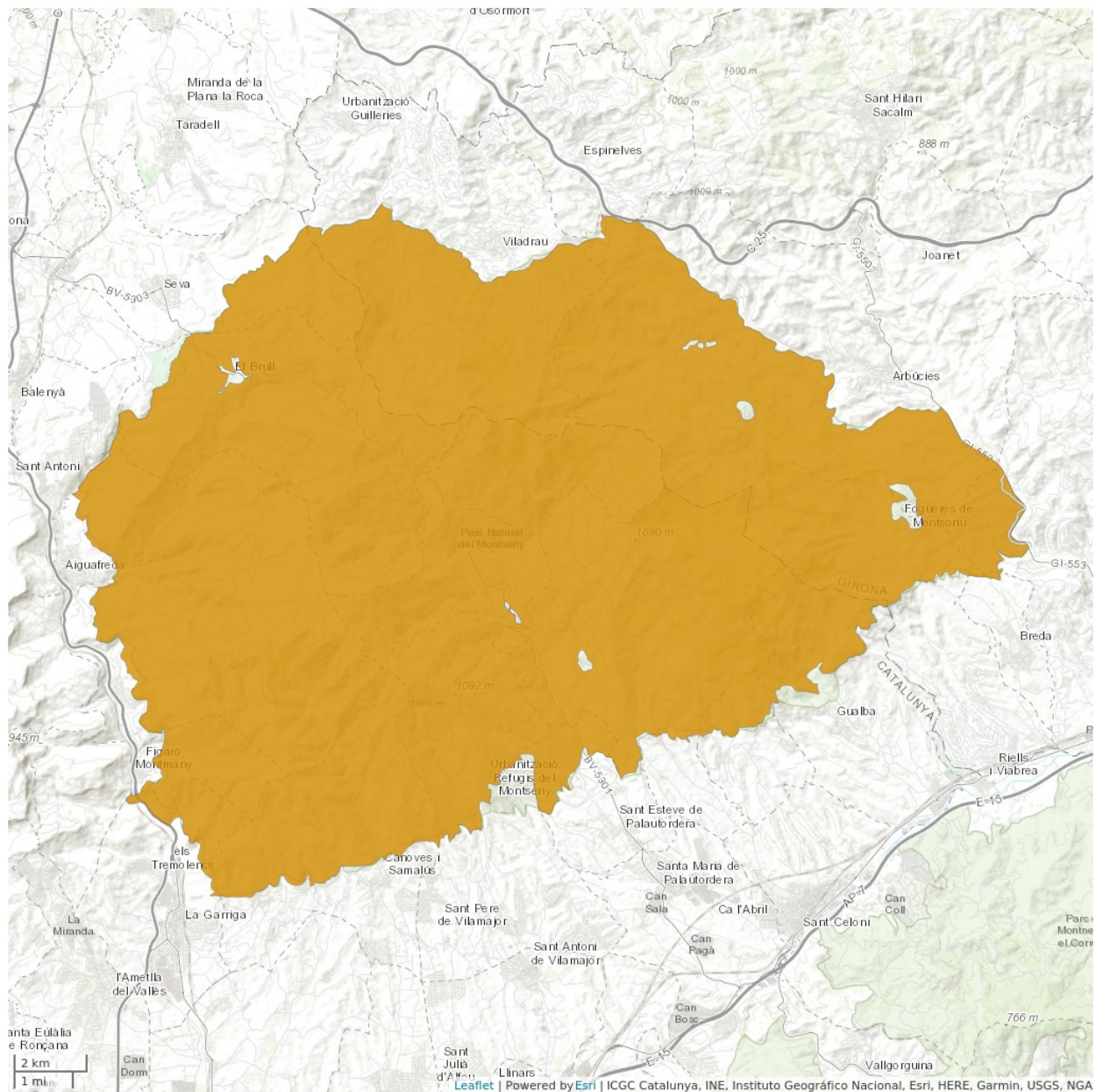
Range Description:

This species is endemic to the El Montseny mountain massif situated in north-eastern Spain. It's only been found in seven streams, which are divided into two groups (one with three subpopulations and one with four), most of which are isolated from each other. All subpopulations occur within the boundaries of El Montseny Natural Park (Carranza and Amat 2005). It is found between 600–1,200 m asl. It occurs in one threat-defined location, and its extent of occurrence (EOO) is 120 km².

Country Occurrence:

Native, Extant (resident): Spain (Spain (mainland))

Distribution Map

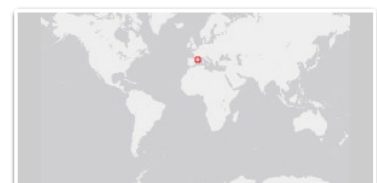
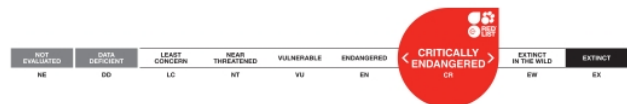


Legend

EXTANT (RESIDENT)

Compiled by:

IUCN (International Union for Conservation of Nature) & Conservation International 2021



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

Its population is thought to number between 1,000–1,500 mature individuals, with an estimated rate of decline of 15% over the last 10 years. Preliminary demographic studies were carried out in two streams (Montori and Campeny 1991); in these streams, this species occurred at very low densities and in one stream it was extinct above 1,000 m asl. The population of this species is divided into two groups that occur on each side of a river valley. These two groups are genetically and morphologically differentiated (Carranza and Amat 2005, Valbuena-Ureña *et al.* 2017b).

The disease chytridiomycosis caused by *Batrachochytrium salamandrivorans* (*Bsal*) is an emerging concern for salamanders in Europe (see Threats section). Extinction risk modelling for *Bsal* calculated the probability of extinction of this species in three generations based on a best and worse case scenario (see model in the attached Supplementary Information). The best case scenario was calculated using the minimum generation length estimate and the worst case scenario using the maximum generation length estimate. For this species the probability of extinction for the best scenario using a generation length of three years is 99.92%, and for the worst scenario using a generation length of six years is 99.99%. Applying criterion E (see spreadsheet; link in the Supplementary Information), this species qualifies for Critically Endangered, as it has a greater than 50% probability of extinction in the wild in three generations. Applying criterion A, the model also predicts that at least 80% or more of the population will be lost within three generation lengths [9–18 years] (see model in the attached Supplementary Information). Due to ongoing decline in the extent and quality of its habitat, the population is already presently decreasing. Subpopulations are declining in the upper parts of all the streams as the water table is being lowered.

For further information about this species, see [Supplementary Material](#).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

This species lives in oligotrophic and cold (below 15°C) streams, with very steep slopes and rocky beds of very cracked rocks, preferentially in beech forest (*Fagus sylvatica*) but also in Holm Oak forest (*Quercus ilex*), with patches of Common Alder *Alnus glutinosa* (Carranza and Amat 2005). Neither juveniles or adults have been found on land (Carranza and Amat 2005). If this behaviour is confirmed, the only way that contact among subpopulations could occur is by moving down one watercourse and then up other tributaries of the same river system, although this is very unlikely. It therefore seems possible that there is a considerable degree of isolation between the subpopulations.

Generation length has been estimated to be similar to its congener, *Calotriton asper*, which ranges between three (at lower elevations) to six (at higher elevations) years (I. Martínez-Solano pers. comm. August 2021).

Systems: Freshwater (=Inland waters)

Use and Trade (see Appendix for additional information)

There are no records of this species being utilized.

Threats (see Appendix for additional information)

The drying out of mountain streams is the most immediate threat to this species. Large amounts of water are being extracted from El Montseny, and this is bottled and sold widely in Spain and Europe. As a result, El Montseny is drying out and the distribution of this species is being reduced dramatically. The low population size of the species could also be due to past human alteration of its original habitat (Carranza and Amat 2005). The species might also be affected by global warming. All future scenarios projected a significant decrease in annual streamflow from 21% to as much as 67% (Ledesma *et al.* 2019) and the frequency and length of low streamflow events will dramatically increase. The beech (*Fagus sylvatica*) forest, an excellent habitat for this species, has shifted upwards by 70 m at the highest altitudes (1,600–1,700 m asl) since 1945, and is being replaced by Holm Oak (*Quercus ilex*) forest at lower altitudes (800–1,400 m) (Peñuelas and Boada 2003). In future scenarios the potential change in vegetation toward an expansion of holm oak forests will be more important than climate changes in determining threshold low flow conditions (Ledesma *et al.* 2019). In other words, the most favoured woodland habitat of the species is moving up into areas where the streams are drying out.

Batrachochytrium salamandrivorans (*Bsal*) is an emerging fungal pathogen affecting amphibians and is closely related to *B. dendrobatidis* (*Bd*), which has had a devastating impact on amphibian populations around the world. Both pathogens cause the infectious disease chytridiomycosis, and both are believed to have originated in Asia (for *Bsal*: Martel *et al.* 2014, for *Bd*: O'Hanlon *et al.* 2018). While *Bd* can cause disease in frogs, salamanders and caecilians, *Bsal* has only been reported to cause disease in salamanders and newts, although the pathogen has been detected in some frog species (Martel *et al.* 2013). *Bd* has spread globally, but at present *Bsal* is only known in Asia and has been introduced to Europe, most likely through the pet trade in salamanders (Martel *et al.* 2014, Nguyen *et al.* 2017). Currently in Europe, *Bsal* has been detected in the wild in the Netherlands, Belgium, Germany (e.g., Thein *et al.* 2020) and Spain (Martel *et al.* 2020), and has been shown to be highly pathogenic to most urodelan taxa in Europe (Martel *et al.* 2014). Laboratory trials for susceptibility to *Bsal* demonstrate that *Calotriton arnoldi* is highly susceptible to the disease, and mortality has been observed in individuals of this species that have been tested (Gilbert *et al.* 2020, F. Pasmans and A. Martel pers. comm. June 2021). With a *Bsal* outbreak in Spain diagnosed at ca. 23 km from its range, this is a major threat to the species. Based on current distribution data for *Bsal*, it is expected that the natural advancement of the infection could be up to 10 km per year which would result in a conservative advancement of 400–500 km per 50 years. Extinction risk modelling (see model in the attached Supplementary Information) has found this species to be at extremely significant risk from *Bsal* in the next immediate three years. Subpopulations have been screened for the presence of *Bd*, with negative results so far (Obón *et al.* 2013).

For further information about this species, see [Supplementary Material](#).

Conservation Actions (see Appendix for additional information)

Conservation Actions In-Place

This species occurs in the El Montseny Natural Park. A captive breeding programme started in 2007 and is being carried out by the Departament de Medi Ambient of the Generalitat de Catalunya, Spain (Valbuena-Ureña *et al.* 2017a). The Life Tritó Montseny project (LIFE15 NATO/SE/000757) started on 1 October 2016 will promote around fifty actions to ensure the conservation of this endemic species and its habitat. The actions will be carried out primarily in the Montseny Natural Park and Biosphere Reserve, as well as in three breeding centres (<http://lifetritomontseny.eu/en>).

Conservation Needed

Management actions are needed, including limiting the expansion of holm oak forest, for ameliorating the species' habitat (I. Martínez-Solano pers. comm. August 2019). There is also an urgent need for a EU (European Union) wide coordinated approach and implementation of the *Bsal* Action Plan by Gilbert *et al.* (2020), which was commissioned by the European Commission.

Research Needed

Research is needed to determine the species' ecological requirements, and to determine management options. Population monitoring will also be important with the risk of *Bsal*.

Credits

Assessor(s): IUCN SSC Amphibian Specialist Group

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Authority/Authorities: IUCN SSC Amphibian Specialist Group

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External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.4. Forest - Temperate	Resident	Suitable	Yes
5. Wetlands (inland) -> 5.1. Wetlands (inland) - Permanent Rivers/Streams/Creeks (includes waterfalls)	Resident	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.3. Unintentional effects: (subsistence/small scale) [harvest]	Past, unlikely to return	-	-	Past impact
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
7. Natural system modifications -> 7.2. Dams & water management/use -> 7.2.6. Abstraction of ground water (commercial use)	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Batrachochytrium salamandrivorans)	Future	Whole (>90%)	Very rapid declines	Medium impact: 7
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
11. Climate change & severe weather -> 11.2. Droughts	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place land/water protection
Occurs in at least one protected area: Yes
In-place species management

Conservation Action in Place
Subject to ex-situ conservation: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action Needed
2. Land/water management -> 2.1. Site/area management
2. Land/water management -> 2.2. Invasive/problematic species control
3. Species management -> 3.4. Ex-situ conservation -> 3.4.1. Captive breeding/artificial propagation
5. Law & policy -> 5.1. Legislation -> 5.1.1. International level
5. Law & policy -> 5.2. Policies and regulations
5. Law & policy -> 5.4. Compliance and enforcement -> 5.4.1. International level

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
3. Monitoring -> 3.1. Population trends

Additional Data Fields

Distribution
Estimated extent of occurrence (EOO) (km ²): 120
Number of Locations: 1
Lower elevation limit (m): 600
Upper elevation limit (m): 1,200
Population
Number of mature individuals: 1,000-1,500
Continuing decline of mature individuals: Yes
Population severely fragmented: No
Continuing decline in subpopulations: Yes

Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Generation Length (years): 3-6
Movement patterns: Not a Migrant

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